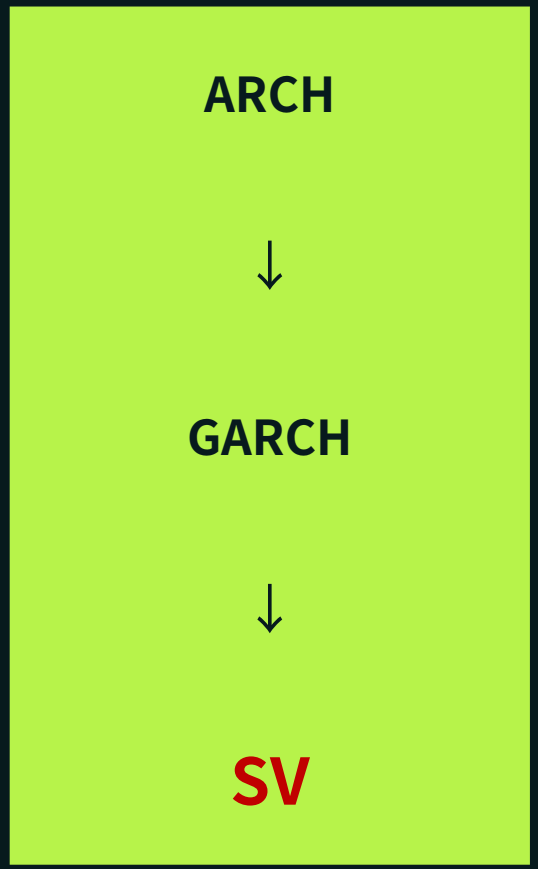
GARCH vs. SV
Fixed or Moving?

Do we already know today's volatility at yesterday's close?



Advanced Financial Time Series

The two models disagree on where volatility comes from

Basic idea — conditional variance: the variance of today's return given information up to yesterday's close

GARCH

Set by returns

- From yesterday's return and variance
- One known number

SV · Stochastic Volatility

Its own shock

- Its own random process
- Uncertain even today

Common ground

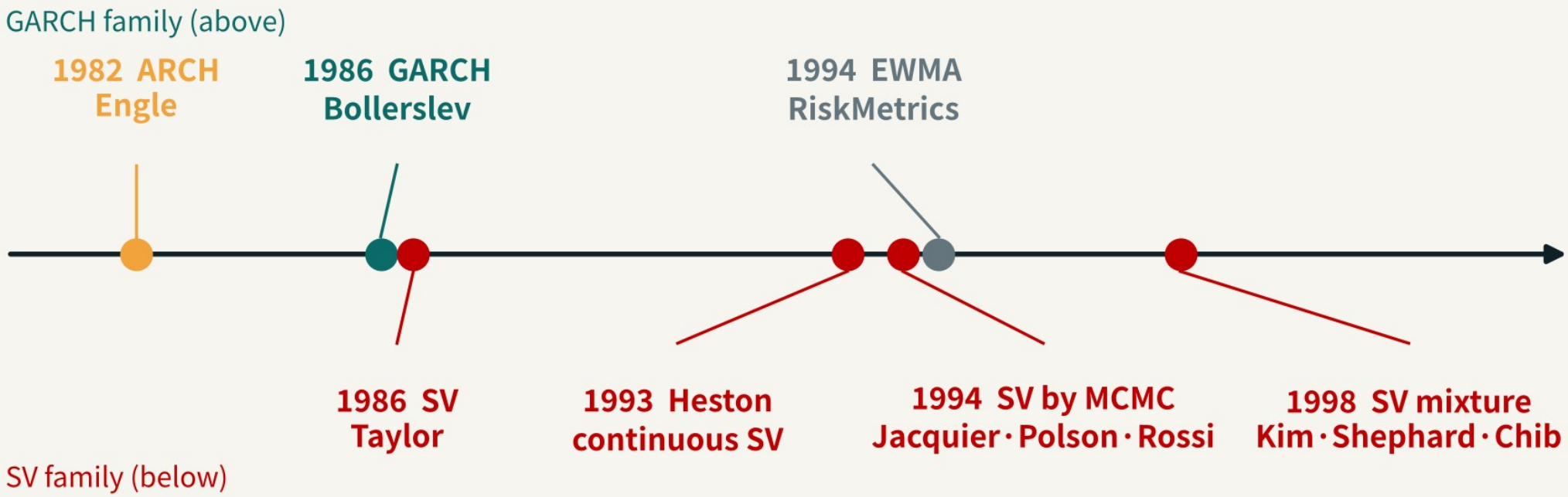
Both explain clustering

- Big moves follow big moves
- About four parameters

Is volatility a function of past data, or a random variable that moves on its own?

Volatility models grew along two branches: GARCH and SV

Timeline — above: observation-driven (GARCH family), below: latent volatility (SV family)



SV appeared the same year as GARCH (1986) but took off only after MCMC in the 1990s

GARCH computes today's variance from yesterday's shock and variance

GARCH(1,1) — parameters μ, α_0, α_1, β_1 (mean, floor, news, memory)

GARCH(1,1)

$$y_t = \mu + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma_t^2)$$
$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

How to read

- ε(t−1): yesterday's shock, observed
- σ^2(t−1): yesterday's variance, computed
- Everything on the right is known
- So today's variance is fixed

GARCH: variance is driven by the observed return ε(t−1)

In SV, log volatility h(t) follows an AR(1) with its own shock η(t)

Standard SV (Taylor, 1986) — parameters μ_h, ϕ, ω_h² (+ mean μ), with h(t) = log σ²(t)

Standard SV

$$y_t = \mu + e^{h_t/2}\, z_t, \qquad z_t \sim N(0,1), \qquad h_t = \log \sigma_t^2$$
$$h_t = \mu_h + \phi\,(h_{t-1} - \mu_h) + \eta_t, \qquad \eta_t \sim N(0, \omega_h^2)$$

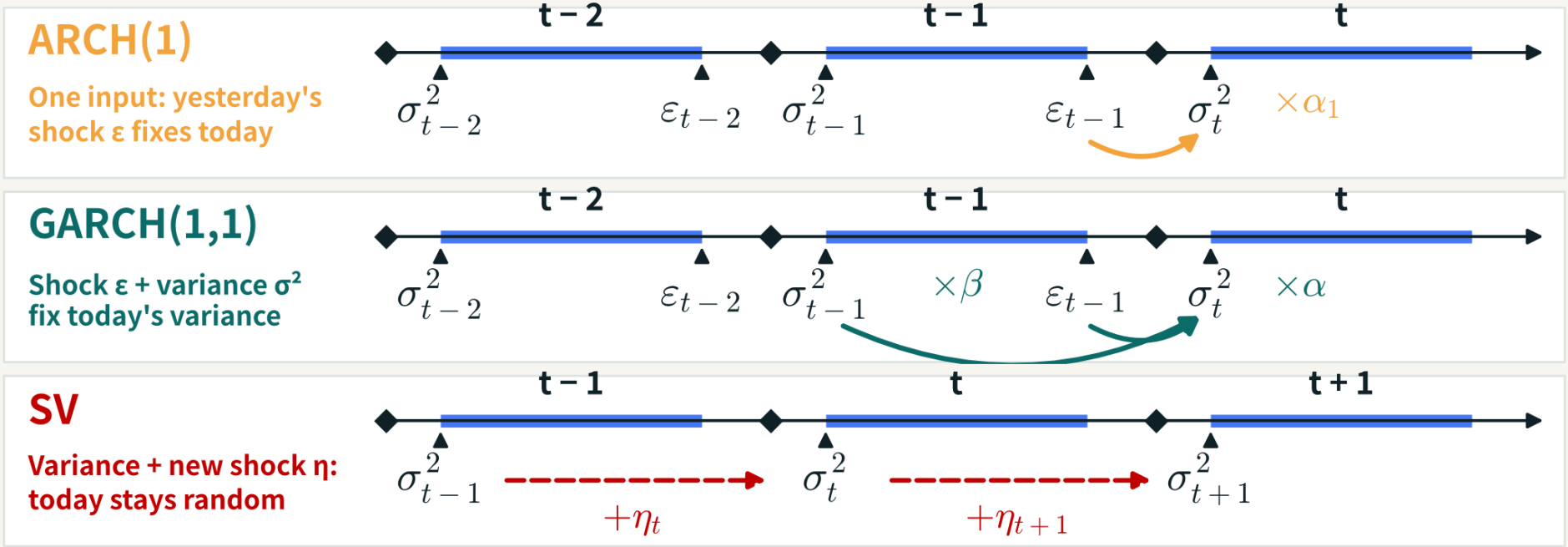
How to read

- Logs keep variance positive
- ϕ: persistence of volatility
- ω_h: volatility of volatility
- η(t) is never observed

SV: variance is driven by a separate, unobserved shock η(t)

On a timeline, GARCH variance is set a day early; SV variance is not

Blue bar = trading day, ◆ = close, short arrow = value known, curve = inputs to today's variance



GARCH: fixed at yesterday's close / SV: today's new shock η keeps it random

In terms of information sets, the difference is a single symbol

Information set F(t−1): all returns observed up to yesterday's close

Information set and today's variance

$$\mathcal{F}_{t-1} = \{\, y_{t-1},\, y_{t-2},\, \dots \,\} \quad \text{(returns observed up to yesterday's close)}$$

$$\text{GARCH}: \quad \sigma_t^2 \in \mathcal{F}_{t-1} \quad \text{(known before today opens)}$$

$$\text{SV}: \quad \sigma_t^2 \notin \mathcal{F}_{t-1} \quad \text{(needs today's new shock } \eta_t\text{)}$$

How to read

- $\in$: computed from yesterday's data
- $\notin$: unknown from yesterday's data alone
- In SV, variance must be inferred

GARCH variance is computed; SV variance is inferred

A simple example: GARCH gives one value, SV gives a range

Source simulation parameters — yesterday's shock ε = 2 (a big drop), yesterday's variance 1

Today's volatility

$$\textbf{GARCH} \quad \alpha_0 = 0.05,\ \alpha_1 = 0.10,\ \beta_1 = 0.85,\ \sigma_{t-1}^2 = 1,\ \varepsilon_{t-1} = 2$$

$$\sigma_t^2 = 0.05 + 0.10 \times 4 + 0.85 \times 1 = 1.30 \ \Rightarrow\ \sigma_t \approx 1.14 \quad (\text{one number})$$

$$\textbf{SV} \quad \mu_h = 0,\ \phi = 0.95,\ \omega_h = 0.25,\ h_{t-1} = 0.20$$

$$h_t = 0.95 \times 0.20 + \eta_t = 0.19 + \eta_t, \qquad 95\%:\ 0.19 \pm 0.49$$

$$\sigma_t = e^{h_t/2} \in [\,e^{-0.15},\ e^{0.34}\,] = [\,0.86,\ 1.40\,] \quad (\text{a range})$$

How to read

- GARCH: 1.14
- SV: 0.86–1.40
- The SV range comes from η

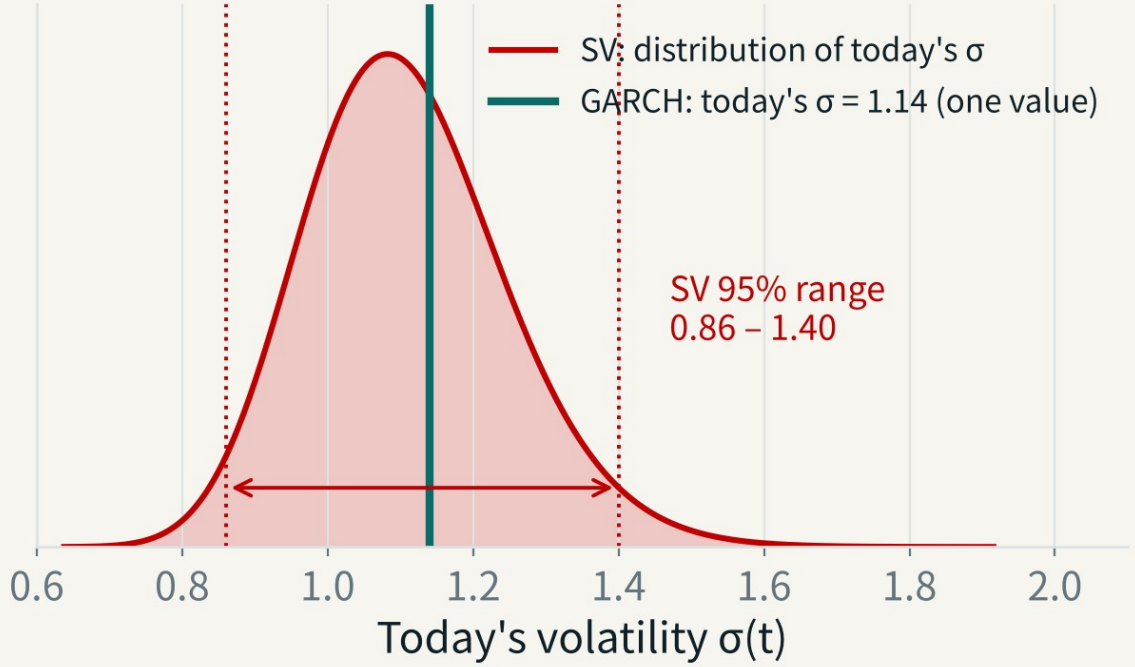
Same morning: GARCH says '1.14', SV says 'roughly 0.86 to 1.40'

In a picture, GARCH is a point and SV is a distribution

Numbers from the previous slide — what each model knows about today's σ(t)

How to read

- Green line: GARCH value
- Red area: SV distribution
- SV prices the risk of volatility itself



SV builds 'volatility of volatility' directly into the model

Estimation: GARCH likelihood is a product, SV's is an integral

If variance is a function of observed data the likelihood is explicit; if latent, it must be integrated out

Likelihood comparison

$$\text{GARCH}: \quad L(\theta) = \prod_t \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{\varepsilon_t^2}{2\sigma_t^2}\right) \quad (\sigma_t^2 \text{ computed recursively})$$

$$\text{SV}: \quad L(\theta) = \int \cdots \int \prod_t p(y_t \mid h_t)\, p(h_t \mid h_{t-1})\, dh_1 \cdots dh_T \quad (T\text{-dim integral})$$

How to read

- GARCH: recursive
- SV: all h paths
- T: sample size

GARCH needs one MLE; SV needs a way around the integral

So GARCH is estimated by MLE, SV by filtering and MCMC

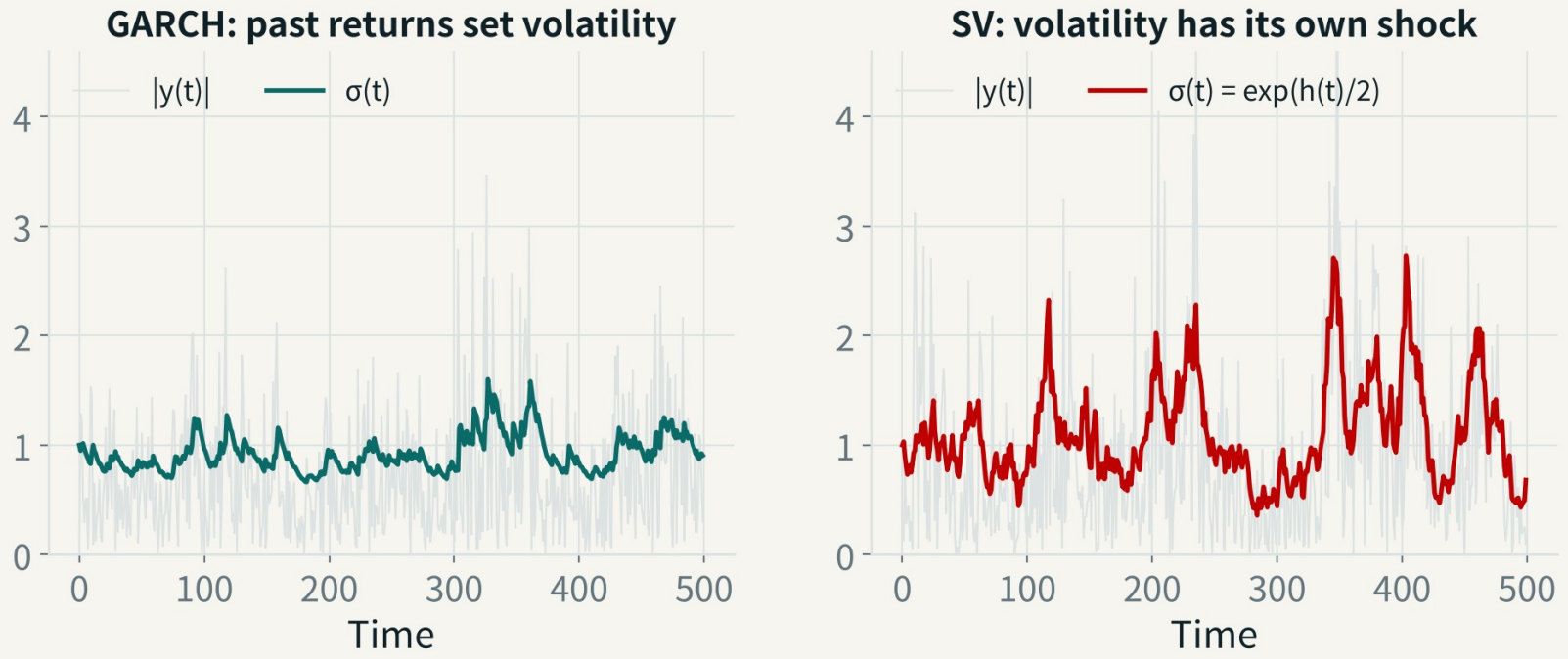
SV route: log-squared returns → normal mixture → Kalman filter or MCMC (Kim, Shephard & Chib, 1998)

	GARCH	SV
Today's variance	**Computed**	**Latent (unobserved)**
Likelihood	**Closed form**	**T-dimensional integral**
Method	**One-shot MLE**	**Kalman filter · MCMC**
Compute time	**Instant**	**Minutes to tens of minutes**
Output	**One variance estimate**	**Posterior of the variance**

SV is more flexible but heavy; GARCH is less flexible but instant

SV describes a world where volatility moves independently of returns

Simulation — GARCH (ω = 0.05, α = 0.10, β = 0.85) vs. SV (ϕ = 0.95, s.d. of η = 0.25)



Left: σ jumps right after a large |y| / Right: σ rises and falls on its own

Use GARCH for fast risk management, SV when volatility risk itself matters

Practical choice — which matters more: speed or uncertainty about volatility?

Situation	Better model	Why
Daily VaR · many assets	**GARCH**	**Instant; today's variance is fixed**
Option pricing · smile	**SV (Heston etc.)**	**Vol of vol creates fat tails**
Volatility uncertainty	**SV**	**Gives a distribution (interval)**
Bayesian · predictive density	**SV**	**Fits naturally with MCMC**
Teaching · quick baseline	**GARCH**	**Simple and easy to read**

Less a contest than a choice between convenience and realism

GARCH: today's variance from yesterday's data

SV: today's variance has a new shock

So GARCH uses MLE, SV uses MCMC

Is volatility fixed or moving?
The answer picks the model and the estimator